About 2,600 results for “review wildfire spread prediction”

Fields of Study | Date Range | Has PDF | Author | Journals & Conferences | More Filters | Filters | Sort by Relevance

Software-Based Simulations of Wildfire Spread and Wind-Fire Interaction

M. Ghodrat | F. Shakeriaski | S. A. Fanaee | A. Simeoni

Environmental Science, Computer Science · Fire · 31 December 2022

TLDR This paper is focused on the investigation of existing simulator models applicable in predicting **wildfire spread** and wind fire interaction and the available software packages are outlined with their broad range of applications in fire dynamic modeling. Expand

11 | PDF | Publisher | Save | Cite

Wildfire Progression Prediction and Validation Using Satellite Data and Remote Sensing in Sonoma, California

Daisy Adhikari | Wan Chen | Yunao Guo | Lilin Huang | J. Gao

Environmental Science, Engineering · International Symposium on Service Oriented… · 1 July 2023

Wildfire has been a serious disaster in the United States for a long time with increasing devastation power, and occurrence risk. Compared to **wildfire** detection in which numerous kinds of research… Expand

3 | IEEE | Save | Cite

Coupling wildfire spread and erosion models to quantify post-fire erosion in Northern Sardinia, Italy

Liliana Del Giudice | B. Arca | +8 authors | M. Salis | Environmental Science · 1 May 2020

<p>High severity **wildfires** can have many negative impacts on ecosystems. In this work, we coupled **wildfire spread** and erosion **prediction** modelling to evaluate the effects of fuel reduction… Expand

Publisher | Save | Cite

A novel wildfire digital-twin framework using interactive wildfire spread simulator

Hyeong-su Kim | Jin-Woo Kwon | Seongjin Yun | Won-Tae Kim

Environmental Science, Computer Science · International Conference on Ubiquitous and Future… · 1 July 2019

TLDR A novel **wildfire** digital-twin framework is proposed using interactive **wildfire spread** simulator to give a **wildfire**-Twin including a **wildfire** simulator the sensing data gathered from oneM2M IoT server and make the digital twin **wildfire prediction**. Expand

17 | IEEE | Save | Cite

Coupling wildfire spread and erosion models to quantify post-fire erosion before and after fuel treatments

M. Salis | L. Giudice | +8 authors | B. Arca | Environmental Science · International journal of wildland fire · 8 October 2019

Wildfires are known to change post-fire watershed conditions such that hillslopes can become prone to increased erosion and sediment delivery. In this work, we coupled **wildfire spread** and erosion… Expand

21 | PDF | Publisher | Save | Cite

A review of machine learning applications in wildfire science and management

P. Jain | Sean C. P. Coogan | Sriram Ganapathi Subramanian | Mark Crowley | Steve Taylor | M. Flannigan

Environmental Science, Computer Science · Environmental Reviews · 2 March 2020

TLDR A scoping **review** of ML applications in **wildfire** science and management and discusses the advantages and limitations of various ML approaches relating to data size, computational requirements, generalizability, and interpretability as well as identifying opportunities for future advances in the science andmanagement of **wildfire**s within a data science context. Expand

530 PDF arXiv Save Cite

FireSight: Utilizing Deep Learning for Wildfire Prediction And Determining Escape Routes

Aniket Mittal Environmental Science, Computer Science · 2023

TLDR Using public data from NASA's MODIS, LP DAAC, University of Idaho, and UC Irvine, 12 input features and 18,545 samples, the fire mask at day t+1 is predicted, and this dataset contains the most variables at 1 km. Expand

1 Save Cite

Sensitivity to the Representation of Wind for Wildfire Rate of Spread: Case Studies with the Community Fire Behavior Model

M. Eghdami P. J. Muñoz Amy L. DeCastro Environmental Science · Fire · 31 March 2025

Accurate **wildfire spread** modeling critically depends on the representation of wind dynamics, which vary with terrain, land cover characteristics, and height above ground. Many fire **spread** models are… Expand

1 Publisher Save Cite

Role of buoyant flame dynamics in wildfire spread

M. Finney Jack D. Cohen +7 authors J. English Environmental Science · Proceedings of the National Academy of Sciences… · 16 July 2015

TLDR New experimental evidence reported here reveals how buoyancy generated by the fire induces vorticity and instabilities in the flame zone that control the convective heating needed to ignite fuel particles and produce **spread**. Expand

264 PDF NAS Save Cite

Application of Artificial Intelligence-Based Regression Methods in the Problem of COVID-19 Spread Prediction: A Systematic Review

J. Musulin Sandi Baressi Šegota +7 authors Elitza Markova-Car Computer Science, Medicine

TLDR A conclusion is drawn from the observation of **review**ed papers that AI-based algorithms have a clear application in COVID-19 epidemiological **spread** modeling and may be a crucial tool in the combat against coming pandemics. Expand

34 PDF Save Cite

← … 10 11 12 13 … →

Stay Connected With Semantic Scholar

Your E-mail Address Sign Up

What Is Semantic Scholar?

Semantic Scholar is a free, AI-powered research tool for scientific literature, based at Ai2.

Learn More

About
About Us
Publishers
Blog
Ai2 Careers

Product
Product Overview
Semantic Reader
Scholar's Hub
Beta Program

API
API Overview
API Tutorials
API Documentation
API Gallery

Research
Publications
Research Careers
Resources

Help
FAQ
Librarians
Tutorials
Contact